

A FAST DOA TRACKING ALGORITHM BASED ON THE EXTENDED KALMAN FILTER

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Abstract. Many direction of arrival (DOA) tracking algorithms suffer heavy computational complexity due to their dependence on the sample covariance matrix. We propose a new DOA tracking algorithm that uses the extended Kalman filter (EKF). Because the proposed algorithm is based on a single snapshot, it requires low computational complexity. Our simulation illustrates that the proposed algorithm performs better than other methods

Key Words: DOA tracking, extended Kalman filter, Antenna array

1. INTRODUCTION

There are many tracking algorithms using a sensor array [?, ?]. Most of the algorithms are based on the covariance made with a block of measurement snapshots obtained from a sensor array. These algorithms suffer two significant drawbacks. The one is that large errors could occur due to the assumption that the trajectory of the DOA is fixed or linear function of time within the block. The other is that the computational complexity of them is very large due to the need of processing a block of measurements at each sampling time.

In this paper, we propose a new DOA tracking algorithm that overcomes these shortcomings using an EKF-based algorithm. The EKF is the most widely used methods for recursive nonlinear estimation [?]. Because it depends on a single snapshot rather than the sample covariance, the required computation is significantly low and the adaptation to moving targets is faster than other ones.

In section 2, the data model is presented and one of the previous algorithms in [?] is explained. The proposed algorithm is explained and its computational complexity is given in section 3. In section 4, we study the performance of the proposed algorithm through simulation. Finally conclusions are made in section 5.

2. PROBLEM FORMULATION

Consider an array of p uniformly-spaced sensors with inter-sensor distance d . We assume that q targets emit narrowband LOS (Line Of Sight) signals. Fading effects, Doppler effects and multipath signals are assumed to be negligible. Then the received waveform on the array can be expressed as

$$z(t) = A(\theta(t))s(t) + n(t) \quad (1)$$

where $z(t) = [z_1(t), z_2(t), \dots, z_p(t)]^T$ denotes the received complex waveform vector, $s(t) = [s_1(t), s_2(t), \dots, s_q(t)]$ denotes the complex signal vector emitted by the targets, $n(t) = [n_1(t), n_2(t), \dots, n_p(t)]^T$ is the complex signal vector of measurement noise and $A(\theta(t)) = [a(\theta_1(t)), a(\theta_2(t)), \dots, a(\theta_q(t))]$ is the $p \times q$ steering vector matrix. The steering vector is given by $a(\theta_i(t)) = [1, \exp(-j2\pi \frac{d}{\lambda} \sin(\theta_i(t))), \dots, \exp(-j2(p-1)\pi \frac{d}{\lambda} \sin(\theta_i(t)))]^T$. The sampling interval is denoted by T and we get the sampled measurement $z(n) = z(nT)$. Then direction-of-arrival (DOA) tracking problem is to estimate $\theta(n) = \theta(nT)$, $n = 1, 2, \dots$, using the measurement snapshots.

Recently, a DOA tracking algorithm based on a modified beamformer has been proposed [?]. The key idea of the algorithm is that the DOA $\theta(t)$ within interval $[t_0, t_0 + MT]$ can be approximated by a local truncated Taylor series $\theta(t_0) + \theta'(t_0)(t - t_0)$ where M is the sliding window size. This approximation will be justified when the window size is sufficiently small and the higher order terms of

Taylor series of $\theta(t)$ is negligible. Then the DOA tracking problem is to estimate the state vector $x = (x_0, x_1)^T$ where the DOA $x_0 = \theta(t_0)$ and the angular velocity $x_1 = \theta'(t_0)$. The derivation is based on the weighted least squares approach. The result is referred to as the local polynomial approximation (LPA)-beamformer whose cost function is

$$f(x, k_0) = \sum_{n=k_0}^{k_0+M} |a^H(x_0 + x_1(n - k_0)T)z(n)|^2. \quad (2)$$

The state vector is obtained by a two-dimensional search of the maxima of the above cost function. When $x_1 = 0$, the LPA-beamformer is the conventional beamformer. The major drawback of the LPA-beamformer is to require heavy computational load and the two-dimensional searching for the maximum of the cost function. The other drawback is that the linear approximation of the DOA cannot provide an accurate estimate when the target is moving fast on a nonlinear track.

3. THE EKF-BASED DOA TRACKING ALGORITHM

Let us define the state by $x_i(k) = [\theta_i(k), \dot{\theta}_i(k)]^T$ for i th target. We assume that the dynamics of the i th target is governed by following state equation.

$$x_i(k+1) = Fx_i(k) + Gw_i(k) \quad (3)$$

$$F = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} \frac{1}{2}T^2 & 0 \\ 0 & T \end{bmatrix}$$

where $w_i(k)$ is the process noise with covariance matrix Q_i . The last term $Gw_i(k)$ explains the random acceleration. The linearized measurement model is obtained as

$$z(k+1) = H(k, s(k))x(k) + v(k) \quad (4)$$

where

$$H(k, s(k)) = \begin{bmatrix} \text{Real}(\frac{\partial}{\partial x}(A(\theta(t))s(t))) \\ \text{Imag}(\frac{\partial}{\partial x}(A(\theta(t))s(t))) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ s_{(1,1)} & 0 & \cdots & s_{(q,1)} & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ s_{(1,p-1)} & 0 & \cdots & s_{(q,p-1)} & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ c_{(1,1)} & 0 & \cdots & c_{(q,1)} & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ c_{(1,p-1)} & 0 & \cdots & c_{(q,p-1)} & 0 \end{bmatrix}$$

is the real-valued Jacobian of $A(\theta(t))s(t)$ in (1), the measurement noise $v(k)$ is assumed to have the known covariance matrix $R = E(v(k)v(k)^T)$ and the state vector $x(k)$ is the stacked column vector of q targets' states $[x_1^T, x_2^T, \dots, x_q^T]^T$, where

$$s_{(i,b)} = -\sin(\pi b \sin(\theta_i(k))) \cos(\theta_i(k))s_i(k)$$

$$c_{(i,b)} = -\cos(\pi b \sin(\theta_i(k))) \cos(\theta_i(k))s_i(k)$$

We will use the following notation in EKF algorithm.

$\hat{x}_i(k|k-1)$ = the predicted estimate of $x_i(k)$ based on $[z(1), \dots, z(k-1)]$

$\hat{x}_i(k|k)$ = the filtered estimate of $x_i(k)$

The symbol $s_i(k)$ in (??) can be estimated by using the ML (maximum likelihood) approach:

$$\hat{s}_i(k) = (A^H(\hat{\theta}_i(k|k-1))A(\hat{\theta}_i(k|k-1)))^{-1} A^H(\hat{\theta}_i(k|k-1))z(k) \quad (5)$$

where $\hat{\theta}_i(k|k-1)$ is the first element of $\hat{x}_i(k|k-1)$.

Then our EKF-based algorithm can be summarized as follows

$$\hat{x}(0|0) = \mathbf{x}_0, \mathbf{P}(0|0) = \mathbf{P}(0)$$

FOR $k = 1, 2, 3, \dots$ **DO**

$$\hat{\mathbf{x}}(k|k-1) = \mathbf{F}\mathbf{x}(k-1|k-1)$$

$$\mathbf{P}(k|k-1) = \mathbf{F}\mathbf{P}(k-1|k-1)\mathbf{F}^T +$$

$$\mathbf{G}\mathbf{Q}\mathbf{G}^T$$

$$\mathbf{R}_e(k) = \mathbf{R} + \mathbf{H}(k)\mathbf{P}(k|k-1)\mathbf{H}^T(k)$$

$$\mathbf{K}(k) = \mathbf{P}(k|k-1)\mathbf{H}^T(k)\mathbf{R}_e^{-1}(k)$$

$$\hat{\mathbf{x}}(k|k) = \hat{\mathbf{x}}(k|k-1) + \mathbf{K}(k)[z(k) - \mathbf{h}(\hat{\mathbf{x}}(k|k-1), \hat{\mathbf{s}}(k))]$$

$$\mathbf{P}(k|k) = \mathbf{P}(k|k-1) - \mathbf{P}(k|k-1)\mathbf{H}^T(k)\mathbf{R}_e^{-1}(k)\mathbf{H}(k)\mathbf{P}(k|k-1)$$

END_FOR

In the above, we have used the notation

$$\mathbf{F} = \text{diag}(\overbrace{F, \dots, F}^q)$$

$$\mathbf{h}(\hat{\mathbf{x}}(k|k-1), \hat{\mathbf{s}}(k)) = A(\hat{\theta}(k|k-1))\hat{\mathbf{s}}(k)$$

Now we address the data association problem. When the targets are well separated, $\theta_i(k|k)$ will be associated to the k th target with no difficulty using the above algorithm. However, when targets are crossing, i.e., $|\hat{\theta}_i(k|k-1) - \hat{\theta}_j(k|k-1)| < \delta$ for some different i, j and some threshold δ , the association will be failed owing to large error in the symbol estimation in (??). If the velocities of the targets are sufficiently different, however, the

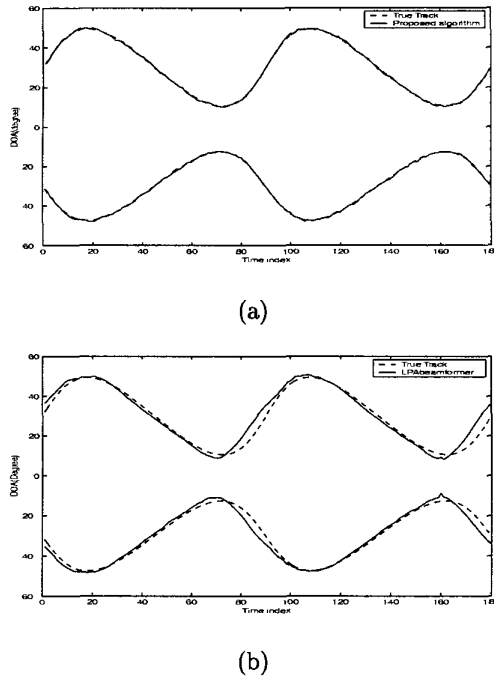


Figure 1: (a) Tracking performance of the proposed algorithm. (b) Tracking performance of the LPA-beamformer

crossing problem can be solved using just the time-update in (??). In this case, tracking algorithm ignores measurements.

We compare the computational complexity of the proposed algorithm with the LPA-beamformer. Assume that q is not large. Proposed algorithm requires the number of real multiplications $5q^3 + 10pq^2 + 8p^2q$. Computational complexity of the LPA-beamformer depends on the range and searching intervals of x_0 and x_1 in (??). If we assume the range and searching intervals of x_0 is $[-90^\circ, 90^\circ]$ and α° and those of x_1 is $[-3^\circ, 3^\circ]$ per sample and β° per sample, then the LPA-beamformer requires the complex multiplications of $108\alpha\beta pM$ and the square root $108\alpha\beta M$ to generate the map of the cost function. The required computation of maximum-search operation of the LPA-beamformer is negligible when q is small. Brief investigation above shows that computational burden is much relieved in the proposed algorithm.

4. SIMULATION RESULTS

We consider a uniform linear array of eight omnidirectional sensors with half wavelength

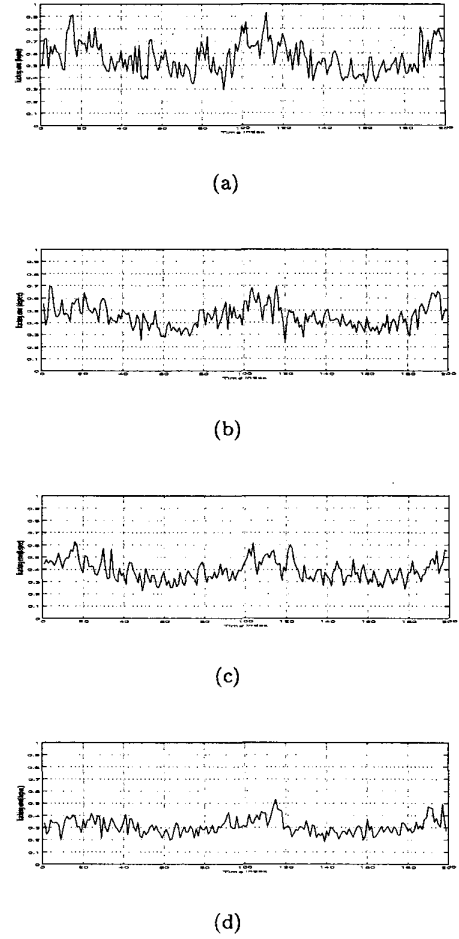


Figure 2: Tracking error vs. time index. (a) SNR = 6dB. (b) SNR = 8dB. (c) SNR = 10dB. (d) SNR = 12dB.

separation. We assume that the targets emit BPSK signals of equal power.

In the first simulation, we compare the proposed algorithm with the LPA-beamformer [?]. The number of targets is two and the SNR (Signal to Noise Ratio) is 6dB. The true initial DOA is $[30^\circ, -30^\circ]$. We set the initial state of EKF by $x(0) = [31^\circ, 0, -29^\circ, 0]$. The measurement noise covariance matrix R is assumed to be known exactly. The process noise covariance matrix Q is set by $0.2I$, and $P(0)$ is $0.1I$ where I is the identity matrix. In our simulation of LPA-beamformer the size M of the sliding window is 16, and the searching intervals are 0.5° and 0.5° per sample. We run Monte-Carlo 10 laps. In Fig.1, the tracking perfor-

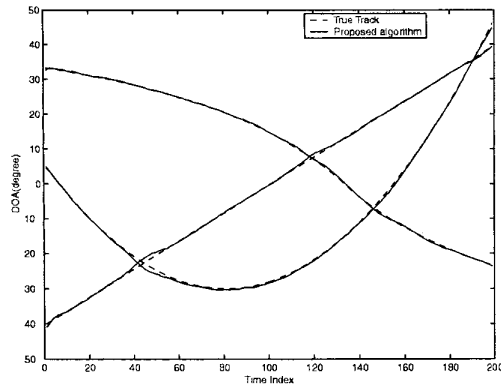


Figure 3: Performance of the proposed algorithm when targets are crossing

performances of the proposed algorithm and the LPA-beamformer are illustrated. As shown in (a), the proposed algorithm tracks the fast varying DOA accurately. In (b) note that the LPA-beamformer experiences some errors most of time. This example reveals the superiority of the proposed algorithm to the LPA-beamformer when targets are moving fast with a nonlinear DOA trajectory.

In the second simulation, performances of the proposed algorithm with different SNRs are illustrated. The simulation environment is the same except that Monte Carlo simulation runs 20 laps. As SNR increases, the overall tracking error of the proposed algorithm is reduced gradually as shown in Fig. 2.

In the third simulation, the tracking performance of the proposed algorithm is illustrated for the crossing trajectories. The sampling time is changed to 0.05s, the threshold δ is 4° and SNR is 6dB. As seen in Fig. 3, the crossing problem can be handled well when the targets are moving with different velocities. When targets are crossing, proposed algorithm performs the time-update only, which will give gradually increasing tracking error.

5. CONCLUSIONS

We present a new DOA tracking algorithm based on the EKF using a single snapshot. The performance of the proposed algorithm has been examined through computer simulation and analysis.

It has been shown that the proposed algorithm has much less computational complexity in com-

parison with the previous algorithm and has good performance even when targets are moving fast. The initialization of the EKF may be done through a conventional DOA estimation method such as MUSIC. The accurate resolution of the crossing problem requires further works.

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